

Adaptive Delta Hedging in Crisis Periods: An Analysis

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Abstract

This report presents a detailed analysis of an adaptive delta-hedging strategy implemented for European options during market crisis periods. Our model uses Black-Scholes to compute greeks, predicts volatility with the GARCH model, and makes use of threshold-based rebalancing with gamma-aware adjustments, accounting for explicit transaction costs and margin constraints. Using synthetic data that mimics adversarial conditions, we evaluate hedging performance during two distinct crisis regimes. We analyse optimal frontiers for gamma and delta rebalancing thresholds and implement a diagnostic framework which provides a reproducible foundation for assessing hedging efficiency under different market regimes.

1 Introduction

Delta hedging is a critical component of any market-making strategy, nullifying exposure to movements in the underlying in order to manage risk. Around the money, hedging is simple, but as the strike moves further from the underlying keeping delta neutral becomes rapidly more difficult. While theoretically exact at arbitrary strike in continuous time under Black-Scholes assumptions, discrete-time implementation introduces several practical challenges: volatility estimation error, transaction costs, liquidity constraints, and market microstructure effects. These challenges become particularly pronounced during crisis periods when market assumptions break down, precisely when hedging operations become most critical.

This report implements and analyses an adaptive delta-hedging strategy specifically designed for crisis conditions, found at [1]. The prototype extends standard approaches through several innovations: dual volatility forecasting with automatic fallback between GARCH and EWMA models and gamma-aware rebalancing that adjusts trade aggressiveness based on curvature risk.

Our primary contribution lies not in developing a deployable trading strategy but in creating a reproducible, modular environment for comparing volatility estimators, thresholding policies, and cost/margin effects during stressed market conditions. This approach provides insights into the fundamental tradeoffs between hedging accuracy, transaction costs, and risk management during periods when standard assumptions fail.

2 Model and Methods

The backbone of every options strategy is of course the Black-Scholes-Merton[2][3] framework, which predicts that the price of a European option satisfies the partial differential equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (1)$$

where V is the option price, S is the underlying price, σ is volatility, and r is the risk-free interest rate. The delta-hedging strategy involves maintaining a position of $\Delta = \frac{\partial V}{\partial S}$ in the underlying asset to neutralize first-order price risk.

Supposing we begin short one option, in discrete time, our portfolio value at time t is given by:

$$\Pi_t = C_t - \Delta_t S_t \quad (2)$$

where C_t is the cash position and $\Delta_t S_t$ is the (value of) underlying we buy while hedging. The tracking error, which measures hedging performance, is defined as:

$$TE_t = (\Pi_t - V_t) - \Pi_0 \quad (3)$$

where V_t is the theoretical option value at time t , and we subtract the initial portfolio value Π_0 to center the error around zero.

The hedging algorithm also implements two volatility forecasting methods depending on how much data is available, providing a robust approach to capturing volatility dynamics. First, the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model provides a sophisticated framework for forecasting that captures both clustering behaviour and fat-tailed return distributions [4][5]. The model specification is:

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad z_t \sim t(\nu) \quad (4)$$

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (5)$$

Here r_t are the returns at t , μ are the average returns, and ϵ_t are the "shocks" at that time. σ_t is then the conditional standard deviation, and z_t is the "innovation", which follows a t-distribution with ν degrees of freedom. ω is the long-run variance and α and β are parameters to fit that govern the evolution of the conditional deviation. We have the parametric restrictions $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$ to ensure positivity of conditional variance, and $\alpha + \beta < 1$ ensures stationarity. We estimate parameters via maximum likelihood, with the log-likelihood function given by:

$$\ell(\theta) = \sum_{t=1}^T \ln f(\epsilon_t | \mathcal{F}_{t-1}) = \sum_{t=1}^T [-\ln(\sigma_t) + \ln g(z_t)] \quad (6)$$

i.e. the log of the likelihoods of each shock given the previous information. The latter expression comes from observing that $f(\epsilon_t | \mathcal{F}_{t-1}) = \frac{g(z_t)}{\sigma_t}$ (σ_t being known using information from t-1 and before). Expanding the expression:

$$\ell(\theta) = \sum_{t=1}^T \left(-\ln(\sigma_t) + \ln \left(\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\nu/2) \sqrt{\pi(\nu-2)}} \right) - \frac{\nu+1}{2} \ln \left(1 + \frac{\epsilon_t^2}{\sigma_t^2(\nu-2)} \right) \right) \quad (7)$$

Minimising this provides us parameters with which to forecast σ_{t+1} as $\sigma_{t+1|t}^2 = \omega + \alpha \epsilon_t^2 + \beta \sigma_t^2$.

However, the GARCH model requires plenty of data, and when we begin hedging (at least in the simulated environment) we do not necessarily have sufficient information to make good use of it. Instead, we use the Exponentially Weighted Moving Average (EWMA), a model which provides a parsimonious alternative for volatility forecasting, placing greater weight on recent observations. The model specification is:

$$\sigma_t^2 = (1 - \lambda) r_{t-1}^2 + \lambda \sigma_{t-1}^2 \quad (8)$$

with decay factor $\lambda = 0.94$ (following table 5.9 in [6]). This recursive formulation can be expanded into an explicit formula:

$$\sigma_t^2 = (1 - \lambda) \sum_{i=0}^{\infty} \lambda^i r_{t-1-i}^2 \quad (9)$$

revealing that the forecast represents an infinite weighted sum of past squared returns, with weights decaying exponentially at rate λ . The half-life of these weights is given by $\ln(0.5)/\ln(\lambda) \approx 11.2$ days for $\lambda = 0.94$, indicating relatively rapid decay and responsiveness to recent market conditions. The EWMA model is particularly advantageous during market regime shifts, as it quickly adapts to changing volatility environments without requiring parameter estimation.

The implementation employs a hierarchical approach: when sufficient data is available (> 30 observations), GARCH(1,1) is preferred for its statistical efficiency and ability to capture volatility persistence. If GARCH estimation fails or insufficient data is available, the system automatically falls back to EWMA, ensuring robustness against model misspecification and data limitations.

When rebalancing, the strategy uses threshold-based triggers with gamma-aware adjustments:

$$\text{Trade} = \begin{cases} \text{Target} - \text{Current Position} & \text{if } |\Delta_{\text{target}} - \Delta_{\text{current}}| > \Delta_{\text{thresh}} \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where the target delta is adjusted based on gamma exposure:

$$\text{Target} = \Delta_{\text{current}} + m(\Gamma) \cdot (\Delta_{\text{pred}} - \Delta_{\text{current}}) \quad (11)$$

with the multiplier $m(\Gamma)$ defined as:

$$m(\Gamma) = \begin{cases} \min\left(2.0, 1.0 + \frac{\Gamma}{\max(\Gamma_{\text{thresh}}, 10^{-12})}\right) & \text{if } \Gamma > \Gamma_{\text{thresh}} \\ 0.8 & \text{otherwise} \end{cases} \quad (12)$$

This approach increases trade aggressiveness when gamma is high, improving responsiveness to curvature risk, while maintaining a conservative adjustment (0.8) while it is low. The alteration of the two threshold values is the most immediately available method of tuning the model. We shall briefly discuss optimality of thresholds later.

The model also incorporates proportional transaction costs [7]:

$$\text{Cost} = c \cdot |\text{Trades}| \cdot S_t \quad (13)$$

where c is the cost parameter (default: 0.1%). A margin constraint also limits cash borrowing:

$$\text{Cash}_t \geq -k \cdot \text{Initial Capital} \quad (14)$$

(though by default we do not have extra cash beyond our short position).

We also generate a variety of diagnostics including time series of price, vol, tracking error, gamma exposure, and correlation between our portfolio value and spot value, as well as statistics such as drawdown and total trade cost. Finally, we report the value at risk (VaR) at α and the expected shortfall (ES) [6][8]. These are, respectively, the α percentile of our losses on any given day, i.e.

$$\text{VaR}_\alpha(L) = \inf\{l \in \mathbb{R} | \mathbb{P}(L \leq l) \geq \alpha\}$$

and the expected loss given loss is already above the VaR, i.e.

$$\text{ES}_\alpha = \mathbb{E}(L | L \geq \text{VaR}_\alpha(L)).$$

Instead of using actual data we prefer synthetic data generated using a discretised form of the Brownian motion:

$$d \log S_t = \mu dt + \sigma_t dW_t + J_t dN_t \quad (15)$$

with σ_t mean-reverting and J_t being superimposed jumps to force crises. This allows us to experiment with a variety of conditions if desired as well as mimic those found in well-known crashes.

3 Experimental Results

We consider two case studies: "2008" and "2020" conditions. It must be emphasised that these are not literally the data from the period but instead similar general conditions. The "2008" conditions are characterised by an extended decline with transient rebounds and vol spikes, and the "2020" by a sharp initial decline and slow but persistent recovery with smaller but consistent volatility in the later period. Full details may be found in the code. The default choices of threshold used in these studies are not necessarily optimal, and are chosen for illustration: $\Delta_{thresh} = 0.005, \Gamma_{thresh} = 0.01$.

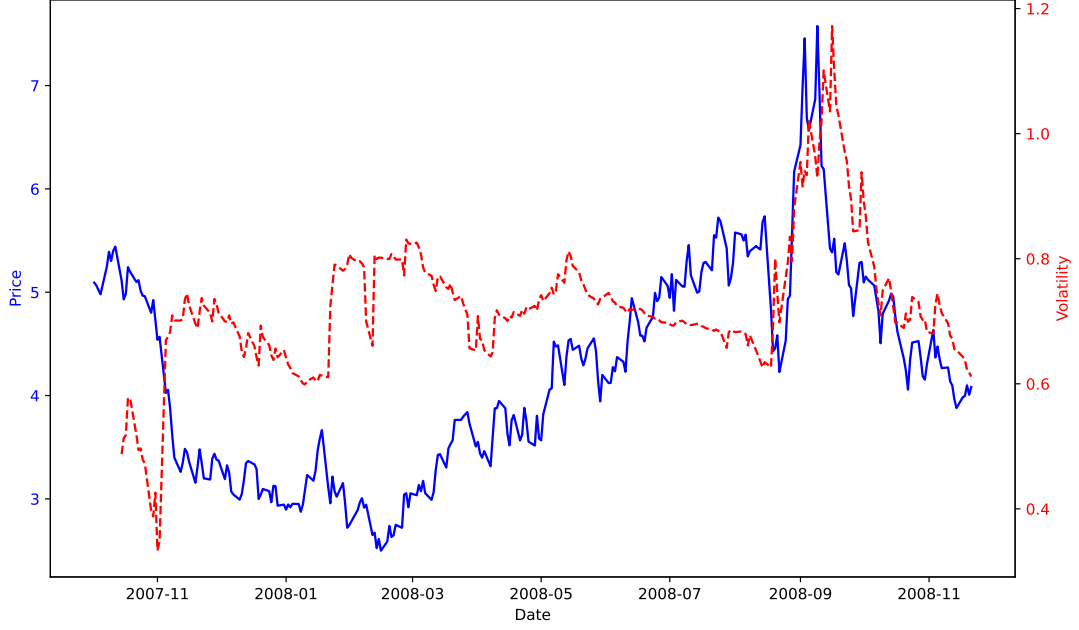


Figure 1: Price and volatility dynamics for the 2008 crisis simulation. Note that despite spikes and an apparent recovery we ultimately see a net loss

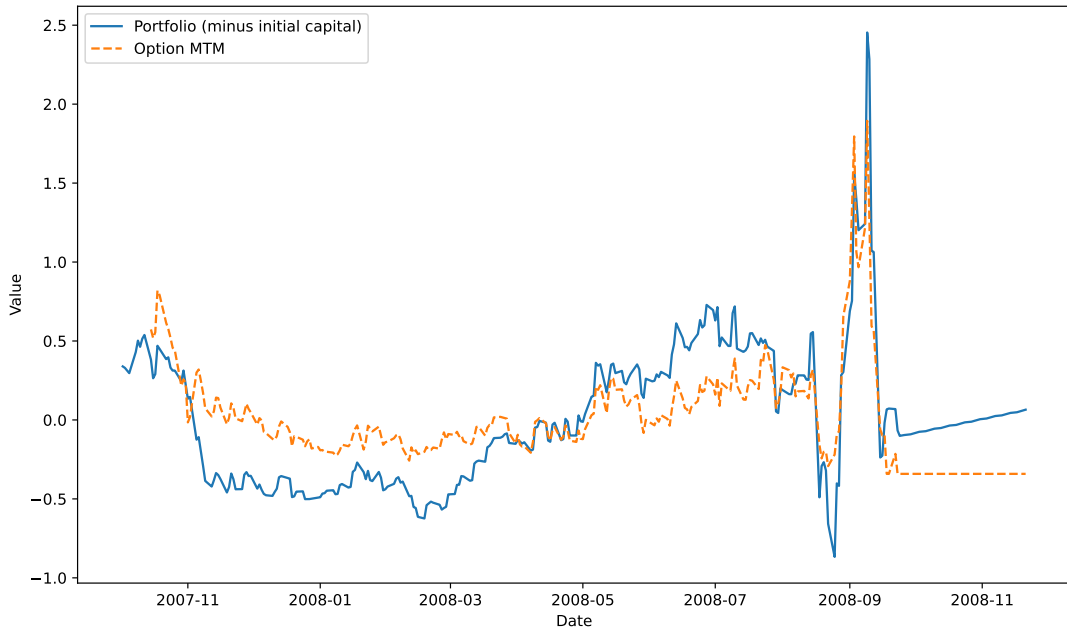


Figure 2: Portfolio value versus option mark-to-market during the 2008 crisis simulation.

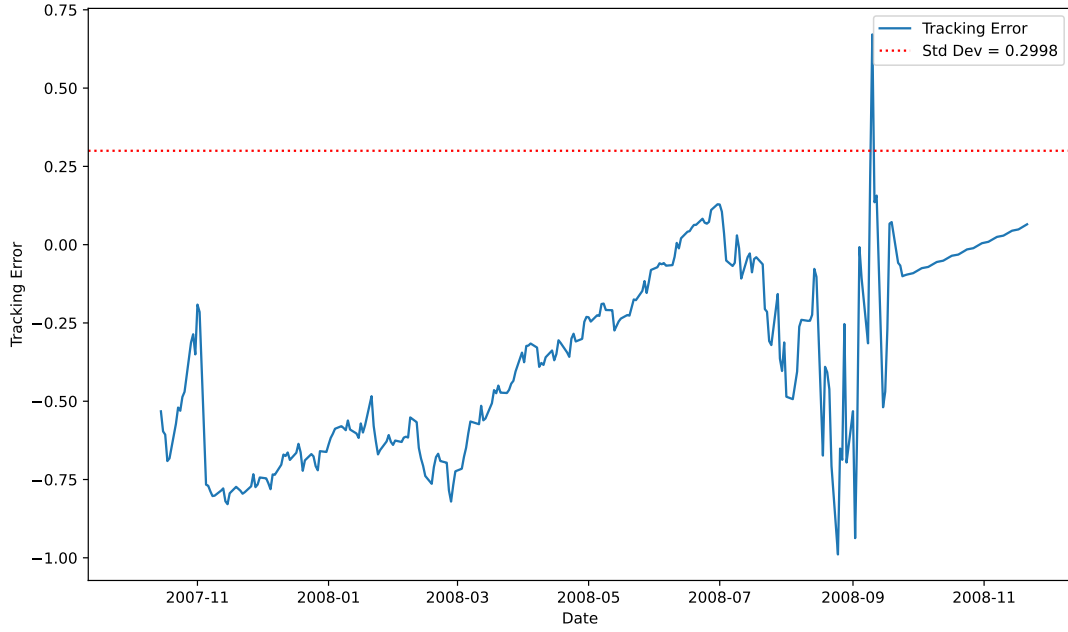


Figure 3: Tracking error time series for the 2008 crisis simulation. Standard deviation: 0.3137.

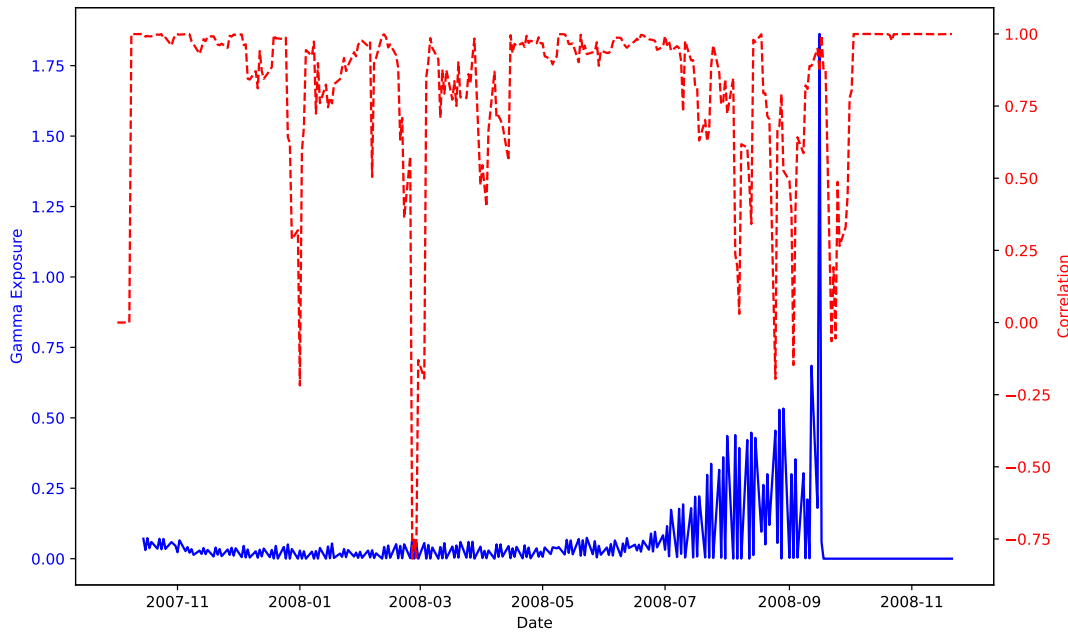


Figure 4: Gamma exposure and daily PnL-spot correlation for the 2008 crisis simulation.

In this situation, the hedger performs reasonably well in the face of adverse conditions, maintaining a low tracking error outside of the most severe spikes, and the ultimate result of its hedges is almost perfectly neutral. Of note is the relatively large PnL-spot correlation through much of the simulation. Though this is not a perfect indicator, it implies a fairly high level of delta leakage (i.e. exposure to underlying) despite the hedging. Of course, various confounders mean this isn't necessarily as bad a sign as it seems. We compare this to the "2020" simulation:

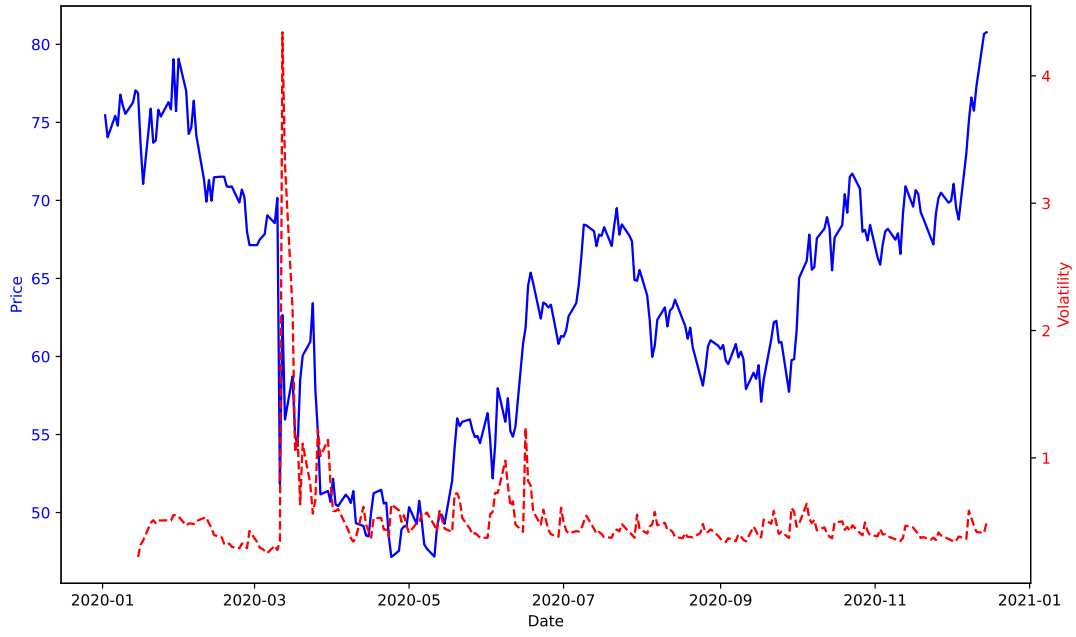


Figure 5: Price and volatility dynamics for the 2020 crisis simulation.

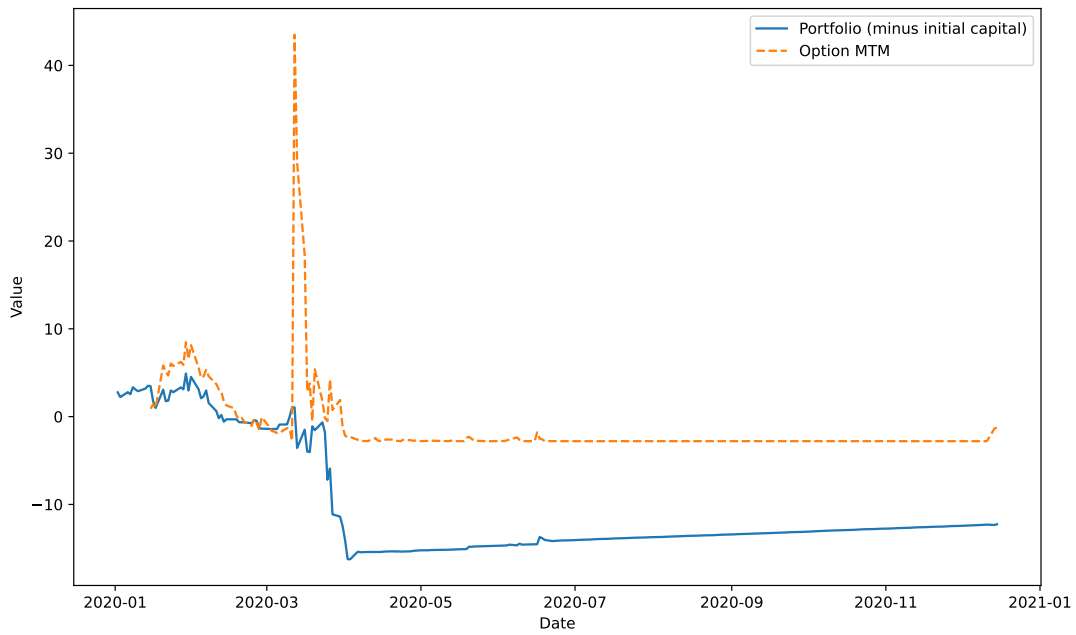


Figure 6: Portfolio value versus option mark-to-market during the 2020 crisis simulation.

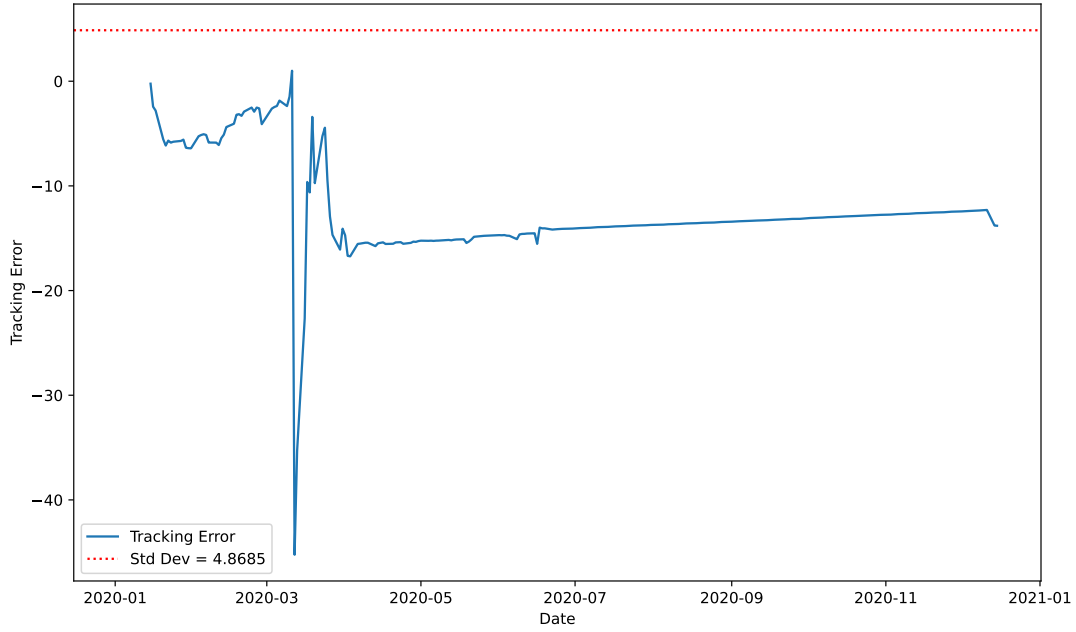


Figure 7: Tracking error time series for the 2020 crisis simulation. Standard deviation: 4.869.

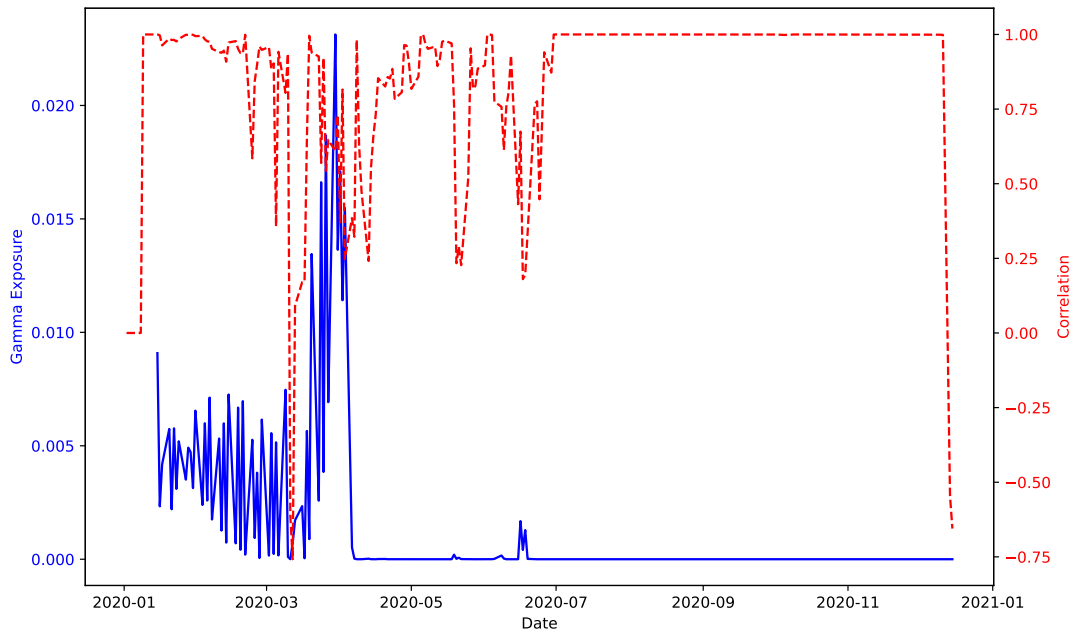


Figure 8: Gamma exposure and PnL-spot correlation for the 2020 crisis simulation.

The hedger performs much worse in this situation, failing to neutralise its initial loss due to a much more tentative trading pattern. The more lasting recovery periods punctuated by sudden downward lurches produce much more tentative behaviour and ensure that for essentially the entire second half of the trading period daily PnL is perfectly correlated with the movements in the underlying. It seems that the choice of threshold here has essentially caused the hedger to fail in its operation. The summary statistics tell the same story:

Table 1: Performance comparison between 2008 and 2020 crisis simulations (note the entries normalised by option value)

Metric	2008 Crisis	2020 Crisis	Difference	% Change
Tracking error std (norm.)	0.876	1.737	0.861	98.3%
Turnover (norm.)	1714	892	-822	-48.0%
Number of trades	250	101	-149	-59.6%
Max drawdown (fraction of IC)	0.050	0.028	-0.22	-44%
95% VaR (norm.)	0.137	1.06	—	—
95% ES (norm.)	0.420	2.42	—	—

(Note that the VaR and ES have not been normalised, as they are more meaningful in raw form, and are thus incommensurable and have not had their changes marked).

We prefer to report error standard deviation over e.g. the mean, since it’s generally centred close to zero and the deviation captures information about how likely error is to cause margin issues. The ”2020” reports greater error deviation, and lower turnover, trade count, and drawdown, to the aforementioned conservative trading. This represents not only a need for better threshold choices but possibly an issue with the fundamental model.

4 Discussion

The implemented dual volatility forecasting approach provides robustness against model misspecification. The GARCH(1,1) model offers superior performance during stable periods by capturing the persistent nature of volatility through its autoregressive structure (i.e. the β) and reaction to recent shocks (α). The t innovation distribution enhances model fit by reflecting the fat-tailed characteristics of financial returns. However, during crisis transitions, the EWMA’s exponential weighting scheme provides faster adaptation to changing volatility regimes without requiring parameter re-estimation. This suggests a more generally hybrid approach might be optimal—using GARCH during stable periods with regime-switching detection to temporarily shift to EWMA during volatility transitions. In our case studies we are essentially in crisis perpetually so this suggestion is less relevant.

The gamma-aware rebalancing strategy provides considerable benefits over fixed threshold approaches. The significantly different trade patterns between the two crises make clear that the thresholds ought to depend on the regime the hedger operates in. Dynamic threshold adjustment is a matter for future work, but we can provide a snapshot of how varying parameters affects the performance of the model:

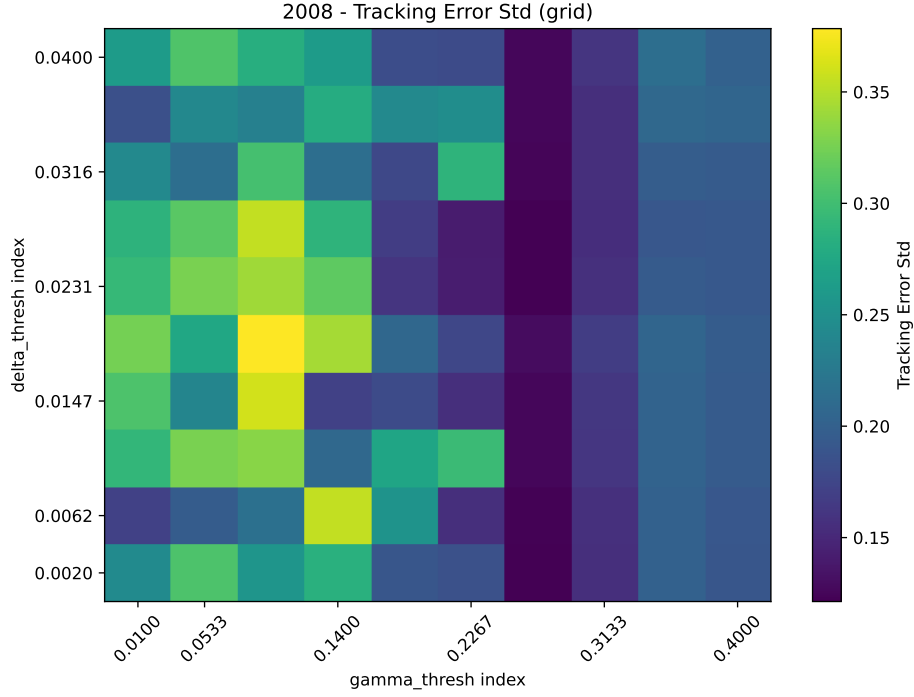


Figure 9: 2008 sensitivity analysis.

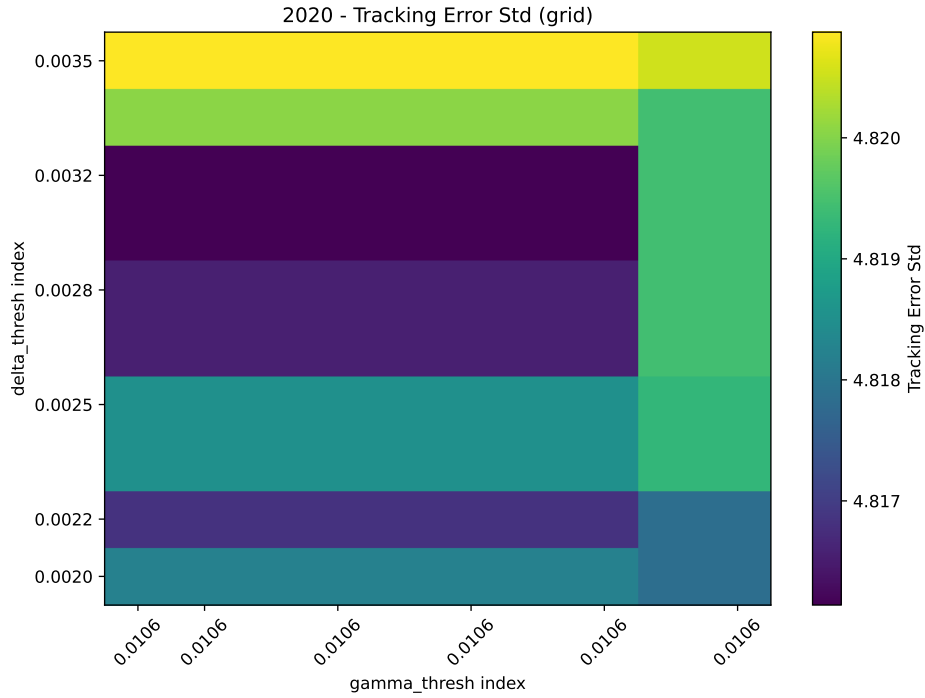


Figure 10: 2020 sensitivity analysis.

You may note that the latter figure is focused on a much smaller region than the former. This is because the initial choice is actually quite close to optimal and we needed to zoom in to see improvement worth noting.

Finally, there is the obligation to discuss the well-known problems with the use of the Black-Scholes model. While it provides a suitable foundation for this analysis, several limitations must

be acknowledged. Firstly, the assumption of constant volatility is particularly problematic during crises when volatility skew becomes pronounced. Our volatility forecasting helps but cannot fully capture the true strike-dependent volatility structure (see e.g. [9],[10]). Of course, our current hedger only works on one strike, but further expansion might see this become an issue. It also fails to account for constraints in liquidity that occur in extreme conditions. The transaction costs are likely underestimated during the sharpest changes. Finally, the supposedly risk-free rate assumption ignores funding spread widening during crises [11], which affects both the discounting and the cost of maintaining hedge positions.

5 Conclusion and Further Work

This report covers the construction and behaviour of an environment in which a delta-hedger can be tested against synthetic and actual data, and considers two case studies. We find that crisis periods present significant challenges for delta hedging, with performance varying dramatically based on crisis dynamics, and that 2020-style crisis simulation showed substantially higher tracking error and risk measures compared to a 2008-style simulation. Trade frequency and patterns differed significantly between crises, and we consider optimality conditions for trading thresholds in both regimes.

For future work, several promising directions emerge:

- Implement explicit regime detection to switch volatility models and hedging parameters based on market conditions.
- Extend our model from single-option hedging to portfolio-level risk management with correlation dynamics.
- Incorporate price-dependent transaction and financing costs to better capture liquidity effects during crises, following the work of e.g. [12].

References

- [1] Apoorv Prabhu. Hedger report, September 2025.
- [2] Fischer Black and Myron S Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, May-June 1973.
- [3] Robert Merton. The theory of rational option pricing. *Bell J Econ Manage Sci*, 4:141–183, 03 1973.
- [4] Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.
- [5] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.
- [6] N.Y.) J.P. Morgan Chase & Co. (New York, J. Longerstaey, Reuters Limited, Reuters Ltd, M. Spencer, Morgan Guaranty Trust Company of New York, and J.P. Morgan. *RiskMetrics: Technical Document*. Guaranty Trust Company of New York, 1996.
- [7] Hayne E Leland. Option pricing and replication with transactions costs. *Journal of Finance*, 40(5):1283–1301, December 1985.
- [8] Carlo Acerbi and Dirk Tasche. Expected shortfall: a natural coherent alternative to value at risk, 2001.

- [9] Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343, 1993.
- [10] Jim Gatheral. *The volatility surface : a practitioner's guide*. Wiley finance series. John Wiley Sons, Hoboken, N.J, 2006.
- [11] Vladimir Piterbarg. Funding beyond discounting: Collateral agreements and derivatives pricing. *Risk Magazine*, pages 97–102, 01 2010.
- [12] Robert Almgren and Neil A Chriss. Optimal execution of portfolio trans-actions. 2000.